इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY
सत्यमेव जयते

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SoCTeamup

Outline

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What is a LEF File?

LEF (Library Exchange Format) is an industry-standard format that describes the abstract physical representation of standard cells and IP blocks

- It provides:
  - ▪ Cell boundary dimensions
  - ▪ Pin geometry and access points
  - ▪ Routing layer usage
  - ▪ Placement grid and alignment rules
  - LEF intentionally avoids detailed polygons to enable fast placement and routing.

Role of the LEF in Physical design?

After synthesis, tools understand logical intent, but not physical feasibility.

- Determine legal placement locations
- Align pins with routing layers
- Enforce row and site compatibility
- Prevent overlaps and routing conflicts

Role of the LEF in Physical design?

LEF ensures:

- Cells are placed without overlaps
- Pins align with metal routing layers
- IP blocks obey floorplan constraints

Without LEF, physical implementation is not possible.

LEF vs. LIB – What's the Difference?

LIB (.lib): Electrical View

- Timing arcs and delays
- Power characterization
- Logical functionality

LEF vs. LIB – What's the Difference?

LIB (.lib): Electrical View

- Geometry and pin placement
- Metal layers and routing access
- Placement grid and symmetry

Both files describe the same cell from orthogonal abstraction domains

Both files describe the same cell but from different perspectives

Key LEF sections: MACROS,SITE,LAYER

- KEY SECTIONS
  - VERSION – Format version
  - UNITS – Distance units
  - LAYER – Metal and via layer definitions
  - SITE – Placement grid info
  - MACRO – Individual cell (like INVX1)
  - OBS: Routing obstructions

Example of LEF snippet INVX1

```
MACRO INVX1
  CLASS CORE ;
  ORIGIN 0 0 ;
  SIZE 1.2 BY 2.0 ;
  PIN A
    DIRECTION INPUT ;
    PORT
      LAYER M1 ;
      RECT 0.1 0.4 0.2 0.6 ;
    END
  END A
  PIN Y
    DIRECTION OUTPUT ;
    PORT
      LAYER M1 ;
      RECT 0.9 1.0 1.0 1.2 ;
    END
  END Y
END INVX1
```

Understanding MACRO Parameters

CLASS	'CORE' for logic gates, 'PAD' for IO cells
SIZE	Width × Height of cell (in microns or units)
ORIGIN	Bottom-left reference corner
PIN	Input or output terminals
LAYER	Which metal layer the pin connects to
RECT	Physical bounding box of the pin

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LEF in floorplanning and placement

SITE defines the cell row height and grid structure

```
SITE coreSite
  SIZE 1.0 BY 2.0 ;
  CLASS CORE ;
  SYMMETRY Y ;
END coreSite
```

- Every standard cell is snapped to a SITE grid for alignment.
- Ensures uniform placement during floorplanning.

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LEF helps physical design tools:

- Place cells along legal rows
- Align pins for proper routing
- Avoid overlap and routing blockages
- Define I/O block locations

LEF vs. DEF – Key Differences

LEF	DEF
Library abstraction	Design instantiation
Reusable across designs	Design-specific
Describes cell geometry	Describes cell placement
Technology view	Implementation view

Both are needed for layout, placement, and routing.

LEF in Open-Source Tools

Tools that use LEF:			
OpenLane	OpenRoad inside openlane	KLayout (for visual LEF/GDSII mapping)	Magic VLSI (LEF-based layout generation)

Students can use LEF files from Google SkyWater130 PDK.

Summary and Best Practices

SUMMARY

- LEF gives abstract physical views of cells
- Used during placement and routing
- Defines geometry, pin location, routing layers
- Complements .lib for complete cell definition

BEST PRACTICES

- Ensure LEF matches .lib in functionality
- Use correct layer/pin sizes for DRC-clean designs
- Learn to read MACRO and SITE sections carefully

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Thank you !